Matrices

Linear Algebra · Lecture 1

1. Matrices and products

Definition 1. Let $m, n \geqslant 1$ be integers. An $m \times n$ **matrix** $A = (a_{ij})$ is a rectangular array of real numbers, its **entries**; a_{ij} stands in row i and column j ($1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$). $\mathbb{R}^{m\times n}$ is the set of all $m \times n$ matrices.

- $A = B$: same size and $a_{ij} = b_{ij}$ for all i, j.
- $\mathbb{R}^n = \mathbb{R}^{n\times 1}$: **column vectors** $x = (x_i)$; $\mathbb{R}^{1\times n}$: **row vectors**. A 1×1 matrix is identified with its entry.
- $\mathrm{col}_j(A) \in \mathbb{R}^m$: column j; $\mathrm{row}_i(A) = (a_{i1}, \ldots, a_{in}) \in \mathbb{R}^{1\times n}$: row i. $a_j = \mathrm{col}_j(A)$, $A = [a_1 \ \cdots \ a_n]$.
- $(A + B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$ ($c \in \mathbb{R}$); $-A = (-1)A$, $A - B = A + (-B)$. The laws of numbers hold, e.g. $A + B = B + A$, $c(A + B) = cA + cB$ (entry by entry).
- 0: all entries 0. $I_n = (\delta_{ij}) \in \mathbb{R}^{n\times n}$, $\delta_{ij} = 1$ ($i = j$), $\delta_{ij} = 0$ ($i \neq j$): the **identity matrix**; its columns $e_1, \ldots, e_n$ are the **unit vectors** of $\mathbb{R}^n$.
- k variables measured on n units: the $n \times k$ **data matrix** (x_{ij}); row i = unit i, column j = variable j, x_{ij} = value of variable j on unit i.

Definition 2. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, the **product** $AB \in \mathbb{R}^{m\times p}$ is

$$(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj} \qquad (1 \leqslant i \leqslant m,\ 1 \leqslant j \leqslant p)$$

AB is defined only when the number of columns of A equals the number of rows of B.

Theorem 3. Let $A \in \mathbb{R}^{m\times n}$.

(a) Whenever the sizes allow the operations: $A(BG) = (AB)G$, $A(B+G) = AB+AG$, $(A+B)G = AG+BG$, $A(cB) = c(AB) = (cA)B$ ($c \in \mathbb{R}$); $I_mA = A = AI_n$, $0A = 0$, $A0 = 0$.

(b) For $M \in \mathbb{R}^{m\times 1}$, $N \in \mathbb{R}^{1\times p}$ and a 1×1 matrix c: the products Mc, cN are the multiples cM, cN, and $McN = c(MN)$.

Proof. (a) Both sides of each rule have the same size: compare the (i, j) entries (Definitions 1, 2). $A(BG) = (AB)G$ for $B \in \mathbb{R}^{n\times p}$, $G \in \mathbb{R}^{p\times q}$:

$$[A(BG)]_{ij} = \sum_{r=1}^{n} a_{ir}(BG)_{rj} = \sum_{r=1}^{n} a_{ir}\Big(\sum_{s=1}^{p} b_{rs}g_{sj}\Big) = \sum_{r=1}^{n}\sum_{s=1}^{p} a_{ir}b_{rs}g_{sj}$$

$$[(AB)G]_{ij} = \sum_{s=1}^{p}(AB)_{is}g_{sj} = \sum_{s=1}^{p}\Big(\sum_{r=1}^{n} a_{ir}b_{rs}\Big)g_{sj} = \sum_{s=1}^{p}\sum_{r=1}^{n} a_{ir}b_{rs}g_{sj}$$

The same np terms $a_{ir}b_{rs}g_{sj}$ in another order: the two entries are equal.
The other rules:

$$[A(B+G)]_{ij} = \textstyle\sum_r a_{ir}(b_{rj}+g_{rj}) = \sum_r a_{ir}b_{rj} + \sum_r a_{ir}g_{rj} = (AB+AG)_{ij}$$
$$[(A+B)G]_{ij} = \textstyle\sum_r (a_{ir}+b_{ir})g_{rj} = \sum_r a_{ir}g_{rj} + \sum_r b_{ir}g_{rj} = (AG+BG)_{ij}$$
$$[A(cB)]_{ij} = \textstyle\sum_r a_{ir}(cb_{rj}) = c\sum_r a_{ir}b_{rj} = [c(AB)]_{ij} = \sum_r (ca_{ir})b_{rj} = [(cA)B]_{ij}$$
$$(I_mA)_{ij} = \textstyle\sum_r \delta_{ir}a_{rj} = a_{ij}, \qquad (AI_n)_{ij} = \sum_r a_{ir}\delta_{rj} = a_{ij} \qquad (\delta_{ir} = 0 \text{ for } r \neq i)$$
$$(0A)_{ij} = \textstyle\sum_r 0 \cdot a_{rj} = 0, \qquad (A0)_{ij} = \sum_r a_{ir} \cdot 0 = 0$$

(b) One term in each sum of Definition 2: $(Mc)_{i1} = m_{i1}c = (cM)_{i1}$, $(cN)_{1j} = cn_{1j}$; so $McN = M(cN) = c(MN)$ by (a). □

By (a), a product of several matrices needs no parentheses. **Powers**: for $A \in \mathbb{R}^{n\times n}$, $A^0 = I_n$ and $A^{k+1} = A^k A$ $(k = 0, 1, 2, \ldots)$.

Theorem 4. Let $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ and $B = [b_1 \ \cdots \ b_p] \in \mathbb{R}^{n\times p}$.

(a) $\mathrm{col}_j(AB) = Ab_j$, that is, $AB = [Ab_1 \ \cdots \ Ab_p]$.

(b) For $x \in \mathbb{R}^n$: $Ax = x_1a_1 + \cdots + x_na_n$. In particular $Ae_j = a_j$.

(c) $\mathrm{row}_i(AB) = \mathrm{row}_i(A)B = a_{i1}\,\mathrm{row}_1(B) + \cdots + a_{in}\,\mathrm{row}_n(B)$.

(d) $(AB)_{ij} = \mathrm{row}_i(A)\,b_j$, and $AB = a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$ (the **outer product expansion**).

Proof. Each part compares entries (Definition 1) with $(AB)_{ij} = \sum_{r=1}^{n} a_{ir}b_{rj}$ (Definition 2).

(a) Fix j; b_j has the entries $b_{1j}, \ldots, b_{nj}$:

$$(Ab_j)_i = \textstyle\sum_r a_{ir}(b_j)_r = \sum_r a_{ir}b_{rj} = (AB)_{ij} \quad (i = 1, \ldots, m) \qquad \text{(Definition 2)}$$
$$\implies Ab_j = \mathrm{col}_j(AB) \qquad \text{(Definition 1)}$$

(b) a_r has the entries $a_{1r}, \ldots, a_{mr}$:

$$(Ax)_i = \textstyle\sum_r a_{ir}x_r = x_1a_{i1} + \cdots + x_na_{in} = (x_1a_1 + \cdots + x_na_n)_i \qquad \text{(Definitions 1, 2)}$$
$$Ae_j = 1 \cdot a_j = a_j \qquad (x = e_j : \ x_j = 1, \ x_r = 0 \text{ for } r \neq j)$$

(c) Fix i; $\mathrm{row}_r(B) = (b_{r1}, \ldots, b_{rp})$:

$$\big(\mathrm{row}_i(A)B\big)_{1j} = \textstyle\sum_r a_{ir}b_{rj} = (AB)_{ij} = \big(\mathrm{row}_i(AB)\big)_{1j} \qquad \text{(Definition 2)}$$
$$\big(a_{i1}\,\mathrm{row}_1(B) + \cdots + a_{in}\,\mathrm{row}_n(B)\big)_{1j} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = (AB)_{ij} \qquad \text{(Definition 1)}$$

(d) $\mathrm{row}_i(A) \in \mathbb{R}^{1\times n}$, $b_j \in \mathbb{R}^{n\times 1}$; $a_r \in \mathbb{R}^{m\times 1}$, $\mathrm{row}_r(B) \in \mathbb{R}^{1\times p}$:

$$\mathrm{row}_i(A)\,b_j = \textstyle\sum_r a_{ir}b_{rj} = (AB)_{ij} \quad (1 \times 1) \qquad \text{(Definition 2)}$$
$$\big(a_r\,\mathrm{row}_r(B)\big)_{ij} = a_{ir}b_{rj} \text{ (one term)} \implies \big(\textstyle\sum_r a_r\,\mathrm{row}_r(B)\big)_{ij} = \sum_r a_{ir}b_{rj} = (AB)_{ij} \qquad \text{(Definitions 1, 2)}$$

□

Example 5. Find AB by entries, by columns, by rows and by outer products, where

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 1 \\ 2 & 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 4}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 2 & -1 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

Solution.

Entries: each is a sum of four products (Definition 2):

$$(AB)_{11} = 1\cdot 1 + 2\cdot 0 + 0\cdot 1 + 1\cdot 2 = 3, \qquad (AB)_{12} = 1\cdot 0 + 2\cdot 1 + 0\cdot 1 + 1\cdot(-1) = 1$$
$$(AB)_{21} = 0\cdot 1 + 1\cdot 0 + 3\cdot 1 + 1\cdot 2 = 5, \qquad (AB)_{22} = 0\cdot 0 + 1\cdot 1 + 3\cdot 1 + 1\cdot(-1) = 3$$
$$(AB)_{31} = 2\cdot 1 + 0\cdot 0 + 1\cdot 1 + 1\cdot 2 = 5, \qquad (AB)_{32} = 2\cdot 0 + 0\cdot 1 + 1\cdot 1 + 1\cdot(-1) = 0$$

$$AB = \begin{pmatrix} 3 & 1 \\ 5 & 3 \\ 5 & 0 \end{pmatrix}$$

Columns: combinations of the columns of A (Theorem 4(a),(b)); rows: combinations of the rows of B (Theorem 4(c)):

$$Ab_1 = 1\begin{pmatrix}1\\0\\2\end{pmatrix} + 0\begin{pmatrix}2\\1\\0\end{pmatrix} + 1\begin{pmatrix}0\\3\\1\end{pmatrix} + 2\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}3\\5\\5\end{pmatrix}$$

$$Ab_2 = 0\begin{pmatrix}1\\0\\2\end{pmatrix} + 1\begin{pmatrix}2\\1\\0\end{pmatrix} + 1\begin{pmatrix}0\\3\\1\end{pmatrix} - 1\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}1\\3\\0\end{pmatrix}$$

$$\begin{aligned}\mathrm{row}_1(AB) &= 1\,(1,0) + 2\,(0,1) + 0\,(1,1) + 1\,(2,-1) = (3,\ 1)\\ \mathrm{row}_2(AB) &= 0\,(1,0) + 1\,(0,1) + 3\,(1,1) + 1\,(2,-1) = (5,\ 3)\\ \mathrm{row}_3(AB) &= 2\,(1,0) + 0\,(0,1) + 1\,(1,1) + 1\,(2,-1) = (5,\ 0)\end{aligned}$$

Outer products: the sum of the four 3×2 matrices $a_r\,\mathrm{row}_r(B)$ (Theorem 4(d)):

$$\begin{aligned}AB &= \begin{pmatrix}1\\0\\2\end{pmatrix}(1,\ 0) + \begin{pmatrix}2\\1\\0\end{pmatrix}(0,\ 1) + \begin{pmatrix}0\\3\\1\end{pmatrix}(1,\ 1) + \begin{pmatrix}1\\1\\1\end{pmatrix}(2,\ -1)\\ &= \begin{pmatrix}1&0\\0&0\\2&0\end{pmatrix} + \begin{pmatrix}0&2\\0&1\\0&0\end{pmatrix} + \begin{pmatrix}0&0\\3&3\\1&1\end{pmatrix} + \begin{pmatrix}2&-1\\2&-1\\2&-1\end{pmatrix} = \begin{pmatrix}3&1\\5&3\\5&0\end{pmatrix}\end{aligned}$$

Example 6. Find NM, MN and X^2, where

$$N = \begin{pmatrix}0&1&0\\0&0&0\\0&0&0\end{pmatrix}, \qquad M = \begin{pmatrix}1&-5&2\\2&3&4\\9&-1&3\end{pmatrix}, \qquad X = \begin{pmatrix}1&1&3\\2&2&6\\-1&-1&-3\end{pmatrix}$$

Solution.

NM by rows (Theorem 4(c)), MN by columns (Theorem 4(a),(b); Theorem 3(a)):

$$\mathrm{row}_1(NM) = 0\,\mathrm{row}_1(M) + 1\,\mathrm{row}_2(M) + 0\,\mathrm{row}_3(M) = \mathrm{row}_2(M), \quad \mathrm{row}_2(NM) = \mathrm{row}_3(NM) = 0$$

$$\mathrm{col}_2(MN) = M\,\mathrm{col}_2(N) = Me_1 = \mathrm{col}_1(M), \quad \mathrm{col}_1(MN) = \mathrm{col}_3(MN) = M0 = 0$$

$$NM = \begin{pmatrix}2&3&4\\0&0&0\\0&0&0\end{pmatrix} \neq MN = \begin{pmatrix}0&1&0\\0&2&0\\0&9&0\end{pmatrix}$$

X^2: $u = \mathrm{col}_1(X)$ has the entries $u_1 = 1$, $u_2 = 2$, $u_3 = -1$; $w_1 = 1$, $w_2 = 1$, $w_3 = 3$ are the entries of $\mathrm{row}_1(X)$:

$$\mathrm{row}_i(X) = u_i\,\mathrm{row}_1(X), \qquad \mathrm{col}_j(X) = w_j u \qquad \text{(read off } X\text{)}$$

$$(X^2)_{ij} = \mathrm{row}_i(X)\,\mathrm{col}_j(X) = \Big(u_i\,\mathrm{row}_1(X)\Big)(w_j u) = u_i w_j\Big(\mathrm{row}_1(X)\,u\Big) \quad \text{(Theorems 4(d), 3(a))}$$

$$\mathrm{row}_1(X)\,u = 1\cdot 1 + 1\cdot 2 + 3\cdot(-1) = 0 \implies X^2 = 0$$

N, M are 3×3 and $NM \neq MN$: in general $AB \neq BA$; square matrices with $AB = BA$ **commute**. $X \neq 0$ and $X^2 = 0$: $AB = 0$ does not force $A = 0$ or $B = 0$ ($A = B = X$), and $AB = AG$ does not force $B = G$ ($A = B = X$, $G = 0$).

2. Transpose, trace and unit vectors

Definition 7. The **transpose** of $A \in \mathbb{R}^{m\times n}$ is $A' \in \mathbb{R}^{n\times m}$, $(A')_{ij} = a_{ji}$: the rows of A' are the transposed columns of A. Column vectors are written $x = (x_1, \dots, x_n)'$. A square A is **symmetric** if $A' = A$.

Theorem 8. For $c \in \mathbb{R}$ and matrices of sizes for which the operations are defined:

$$(A')' = A, \quad (A+B)' = A' + B', \quad (cA)' = cA', \quad (AB)' = B'A', \quad (ABG)' = G'B'A'$$

For $x, y \in \mathbb{R}^n$: $x'y = y'x = \sum_{i=1}^n x_i y_i$, $x'x = \sum_{i=1}^n x_i^2$. $I_n' = I_n$. For $B \in \mathbb{R}^{m\times n}$, $B'B$ and BB' are symmetric.

Proof. Compare entries, with $(A')_{ij} = a_{ji}$ (Definitions 7, 1): $((A')')_{ij} = (A')_{ji} = a_{ij}$, $((A+B)')_{ij} = a_{ji} + b_{ji} = (A'+B')_{ij}$, $((cA)')_{ij} = ca_{ji} = (cA')_{ij}$.
$A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$: $(AB)'$, $B'A' \in \mathbb{R}^{p\times m}$; $1 \leqslant k \leqslant p$, $1 \leqslant i \leqslant m$:

$$(B'A')_{ki} = \textstyle\sum_j (B')_{kj}(A')_{ji} = \sum_j b_{jk}a_{ij} = \sum_j a_{ij}b_{jk} = (AB)_{ik} = ((AB)')_{ki} \quad \text{(Definition 2)}$$
$$(ABG)' = ((AB)G)' = G'(AB)' = G'B'A' \quad \text{(the line above, twice)}$$

$x' \in \mathbb{R}^{1\times n}$ has the entries $(x')_{1i} = x_i$:

$$x'y = \textstyle\sum_i x_i y_i = \sum_i y_i x_i = y'x \quad (1\times 1), \qquad x'x = \sum_i x_i^2 \quad \text{(Definition 2)}$$
$$(I_n')_{ij} = \delta_{ji} = \delta_{ij} \implies I_n' = I_n$$
$$(B'B)' = B'(B')' = B'B, \qquad (BB')' = (B')'B' = BB' \quad \text{(the rules above)}$$

□

Definition 9. The **trace** of $A \in \mathbb{R}^{n\times n}$ is $\operatorname{tr} A = a_{11} + a_{22} + \cdots + a_{nn}$.

Theorem 10. (a) $\operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B$, $\operatorname{tr}(cA) = c \operatorname{tr} A$, $\operatorname{tr}(A') = \operatorname{tr} A$ ($A, B \in \mathbb{R}^{n\times n}$, $c \in \mathbb{R}$).

(b) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ ($A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$).

(c) $\operatorname{tr}(A'A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$ ($A \in \mathbb{R}^{m\times n}$); so $\operatorname{tr}(A'A) = 0 \Rightarrow A = 0$.

Proof. (a) The ith diagonal entries of $A+B$, cA, A' are $a_{ii} + b_{ii}$, ca_{ii}, a_{ii} (Definitions 1, 7); sum over i (Definition 9).

(b) $AB \in \mathbb{R}^{m\times m}$, $BA \in \mathbb{R}^{n\times n}$; Definitions 9, 2:

$$\operatorname{tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ji}, \qquad \operatorname{tr}(BA) = \sum_{j=1}^n (BA)_{jj} = \sum_{j=1}^n \sum_{i=1}^m b_{ji}a_{ij}$$

The same mn terms $a_{ij}b_{ji}$: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

(c) $A'A \in \mathbb{R}^{n\times n}$; Definitions 2, 7:

$$(A'A)_{jj} = \textstyle\sum_{i=1}^m (A')_{ji}a_{ij} = \sum_{i=1}^m a_{ij}^2 \implies \operatorname{tr}(A'A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \quad \text{(Definition 9)}$$
$$a_{kl} \neq 0 \text{ for some } k, l \implies \operatorname{tr}(A'A) \geqslant a_{kl}^2 > 0 \quad (a_{ij}^2 \geqslant 0)$$

So $\operatorname{tr}(A'A) = 0$ forces every $a_{ij} = 0$, that is, $A = 0$.

□

Proposition 11. Let $A, B \in \mathbb{R}^{m\times n}$; e_i is a unit vector of $\mathbb{R}^m$ or of $\mathbb{R}^n$, as the product requires.

(a) $e_i'e_k = \delta_{ik}$, $Ae_j = \mathrm{col}_j(A)$, $e_i'A = \mathrm{row}_i(A)$, $e_i'Ae_j = a_{ij}$.

(b) $A = B \iff Ax = Bx$ for every $x \in \mathbb{R}^n$.

Proof.

(a) The term $(e_i)_r(e_k)_r$ is 1 if $r = i = k$, 0 otherwise:

$$\begin{aligned} e_i'e_k &= \textstyle\sum_r (e_i)_r(e_k)_r = \delta_{ik} && \text{(Theorem 8)} \\ Ae_j &= \mathrm{col}_j(A) && \text{(Theorem 4(b))} \\ e_i'A &= (e_i)_1\,\mathrm{row}_1(A) + \cdots + (e_i)_m\,\mathrm{row}_m(A) = \mathrm{row}_i(A) && \text{(Theorem 4(c))} \\ e_i'Ae_j &= e_i'\,\mathrm{col}_j(A) = a_{ij} && \text{(the line above, for } \mathrm{col}_j(A) \in \mathbb{R}^{m\times 1}\text{)} \end{aligned}$$

(b) $A = B \Rightarrow Ax = Bx$. Conversely, take $x = e_j$ $(j = 1, \ldots, n)$:

$$\mathrm{col}_j(A) = Ae_j = Be_j = \mathrm{col}_j(B) \quad (j = 1, \ldots, n) \implies A = B \qquad \text{((a); Definition 1)}$$

□

3. Diagonal and triangular matrices; the centering matrix

Definition 12. For $d_1, \ldots, d_n \in \mathbb{R}$, $\mathrm{diag}(d_1, \ldots, d_n) \in \mathbb{R}^{n\times n}$ has the diagonal entries $d_1, \ldots, d_n$ and all other entries 0: a **diagonal** matrix. A square A is **upper triangular** if $a_{ij} = 0$ for $i > j$, **lower triangular** if $a_{ij} = 0$ for $i < j$.

Proposition 13. Let $D = \mathrm{diag}(d_1, \ldots, d_n)$ and $F = \mathrm{diag}(f_1, \ldots, f_n)$.

(a) For A with n rows: $(DA)_{ij} = d_i a_{ij}$, $\mathrm{row}_i(DA) = d_i\,\mathrm{row}_i(A)$. For A with n columns: $(AD)_{ij} = a_{ij}d_j$, $\mathrm{col}_j(AD) = d_j\,\mathrm{col}_j(A)$.

(b) $DF = \mathrm{diag}(d_1f_1, \ldots, d_nf_n)$.

(c) If $A, B \in \mathbb{R}^{n\times n}$ are upper (lower) triangular, then AB is upper (lower) triangular, and $(AB)_{ii} = a_{ii}b_{ii}$.

(d) $D^r = \mathrm{diag}(d_1^r, \ldots, d_n^r)$ $(r = 0, 1, 2, \ldots;\ t^0 = 1$ for every $t \in \mathbb{R}$, also $t = 0)$.

Proof.

(a) The (i, r) entry of D is d_i if $r = i$, 0 if $r \neq i$: one term of each sum survives (Definition 2):

$$\begin{aligned} (DA)_{ij} = \textstyle\sum_r D_{ir}a_{rj} = d_ia_{ij} &\implies \mathrm{row}_i(DA) = d_i\,\mathrm{row}_i(A) \\ (AD)_{ij} = \textstyle\sum_r a_{ir}D_{rj} = a_{ij}d_j &\implies \mathrm{col}_j(AD) = d_j\,\mathrm{col}_j(A) \end{aligned}$$

(b) By (a):

$$(DF)_{ij} = d_iF_{ij} = \begin{cases} d_if_i & (i = j) \\ 0 & (i \neq j) \end{cases} \implies DF = \mathrm{diag}(d_1f_1, \ldots, d_nf_n)$$

(c)

- Upper triangular: $a_{ik} = 0$ for $k < i$, $b_{kj} = 0$ for $k > j$; so $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=i}^{j} a_{ik}b_{kj}$ (Definition 2).
- $i > j$: no k with $i \leqslant k \leqslant j$, so $(AB)_{ij} = 0$. $i = j$: only $k = i$, so $(AB)_{ii} = a_{ii}b_{ii}$.
- Lower triangular: only the terms with $j \leqslant k \leqslant i$ remain; $(AB)_{ij} = 0$ for $i < j$, $(AB)_{ii} = a_{ii}b_{ii}$.

(d) Induction on r, with the powers after Theorem 3:

$$D^0 = I_n = \operatorname{diag}(1,\dots,1) = \operatorname{diag}(d_1^0,\dots,d_n^0)$$
$$D^{r+1} = D^r D = \operatorname{diag}(d_1^r,\dots,d_n^r)\operatorname{diag}(d_1,\dots,d_n) = \operatorname{diag}(d_1^{r+1},\dots,d_n^{r+1}) \quad \text{(induction hypothesis; (b))}$$

□

Definition 14. The **summing vector** $\mathbf{1}_n \in \mathbb{R}^n$ has all entries 1; $J_n = \mathbf{1}_n\mathbf{1}_n' \in \mathbb{R}^{n\times n}$ (all entries 1), $\bar{J}_n = \frac{1}{n}J_n$, and $C_n = I_n - \bar{J}_n$, the **centering matrix**. Subscripts are dropped when the size is clear. A square A with $A^2 = A$ is **idempotent**.

Proposition 15. Let $\mathbf{1} = \mathbf{1}_n$, $J = J_n$, $\bar{J} = \bar{J}_n$, $C = C_n$, $I = I_n$. Then $\mathbf{1}'\mathbf{1} = n$, $J^2 = nJ$; $\bar{J}$ and C are symmetric and idempotent: $\bar{J}' = \bar{J} = \bar{J}^2$, $C' = C = C^2$; $C\mathbf{1} = 0$, $\operatorname{tr}\bar{J} = 1$ and $\operatorname{tr} C = n-1$.

Proof.

$\mathbf{1}$, J, $\bar{J}$:

$$\mathbf{1}'\mathbf{1} = \textstyle\sum_{i=1}^n 1\cdot 1 = n \qquad \text{(Theorem 8)}$$
$$J^2 = (\mathbf{1}\mathbf{1}')(\mathbf{1}\mathbf{1}') = \mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = n\,\mathbf{1}\mathbf{1}' = nJ, \qquad \bar{J}^2 = \tfrac{1}{n^2}J^2 = \tfrac{1}{n}J = \bar{J} \qquad \text{(Theorem 3(a),(b))}$$
$$J' = (\mathbf{1}\mathbf{1}')' = (\mathbf{1}')'\mathbf{1}' = \mathbf{1}\mathbf{1}' = J, \qquad \bar{J}' = \tfrac{1}{n}J' = \bar{J} \qquad \text{(Theorem 8)}$$

$C = I - \bar{J}$:

$$C' = I' - \bar{J}' = I - \bar{J} = C \qquad \text{(Theorem 8)}$$
$$C^2 = I\cdot I - I\bar{J} - \bar{J}I + \bar{J}^2 = I - 2\bar{J} + \bar{J} = C \qquad \text{(Theorem 3(a); } \bar{J}^2 = \bar{J})$$
$$C\mathbf{1} = \mathbf{1} - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1} - \tfrac{1}{n}\,n\,\mathbf{1} = 0 \qquad \text{(Theorem 3(a),(b))}$$
$$\operatorname{tr} I = \operatorname{tr} J = n \implies \operatorname{tr}\bar{J} = \tfrac{1}{n}n = 1, \quad \operatorname{tr} C = n-1 \qquad \text{(Definition 9, Theorem 10(a))}$$

□

Example 16. Let $x = (x_1,\dots,x_n)' \in \mathbb{R}^n$, $\bar{x} = \frac{1}{n}\mathbf{1}'x$ (the **mean**) and $C = C_n$. Show that $Cx = x - \bar{x}\mathbf{1} = (x_1-\bar{x},\dots,x_n-\bar{x})'$ (the **deviations** from the mean) and $\sum_{i=1}^n (x_i-\bar{x})^2 = x'x - n\bar{x}^2 = x'Cx$.

Solution.

Deviations:

$$\mathbf{1}'x = x_1 + \cdots + x_n = n\bar{x} \qquad \text{(Theorem 8)}$$
$$Cx = (I - \tfrac{1}{n}\mathbf{1}\mathbf{1}')x = x - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'x) = x - \bar{x}\mathbf{1} = (x_1-\bar{x},\dots,x_n-\bar{x})' \qquad \text{(Theorem 3(a),(b))}$$

Sum of squares; $x'\mathbf{1} = \mathbf{1}'x = n\bar{x}$ (Theorem 8):

$$x'Cx = x'x - \tfrac{1}{n}(x'\mathbf{1})(\mathbf{1}'x) = \textstyle\sum_i x_i^2 - \frac{1}{n}(n\bar{x})(n\bar{x}) = \sum_i x_i^2 - n\bar{x}^2 \qquad \text{(Theorem 3)}$$
$$\textstyle\sum_i (x_i-\bar{x})^2 = \sum_i x_i^2 - 2\bar{x}\sum_i x_i + n\bar{x}^2 = \sum_i x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2$$
$$\textstyle\sum_i (x_i-\bar{x})^2 = x'x - n\bar{x}^2 = x'Cx \qquad (x'x = \sum_i x_i^2\text{; the two lines above})$$

Example 17. Four units ($n = 4$) are measured on two variables ($k = 2$):

$$X_0 = (x_{ij}) = \begin{pmatrix} 2 & 3 \\ 0 & 3 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4\times 2}$$

With $C = C_4$, find the **mean vector** $\bar{x} = (\bar{x}_1, \bar{x}_2)'$ of the column means $\bar{x}_j = \frac{1}{n}\sum_i x_{ij}$, the **centered data matrix** CX_0 and the matrix $X_0'CX_0$ of **corrected sums of squares and products**.

Solution.

Mean vector and CX_0: Theorem 4(a) and Example 16 for the columns:

$$\mathbf{1}'X_0 = \left(\textstyle\sum_i x_{i1},\ \sum_i x_{i2}\right) = (4,\ 8) = 4\bar{x}', \qquad \bar{x} = \tfrac{1}{4}X_0'\mathbf{1} = (1,\ 2)' \quad \text{(Definition 2, Theorem 8)}$$

$$\mathrm{col}_j(CX_0) = C\,\mathrm{col}_j(X_0) = \text{the deviations of variable } j \text{ from } \bar{x}_j$$

$$\mathbf{1}'X_0 = (X_0'\mathbf{1})' = n\bar{x}' \implies CX_0 = X_0 - \tfrac{1}{n}\mathbf{1}(\mathbf{1}'X_0) = X_0 - \mathbf{1}\bar{x}' \qquad \text{(Theorems 8, 3)}$$

$$CX_0 = \begin{pmatrix} 2-1 & 3-2 \\ 0-1 & 3-2 \\ 2-1 & 1-2 \\ 0-1 & 1-2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$X_0'CX_0$: $C = C^2 = C'C$ (Proposition 15):

$$X_0'CX_0 = X_0'C'CX_0 = (CX_0)'(CX_0) \qquad \text{(Theorem 8)}$$

$$\left((CX_0)'(CX_0)\right)_{jl} = \mathrm{col}_j(CX_0)'\,\mathrm{col}_l(CX_0) \qquad \text{(Theorem 4(d))}$$

$$= 1+1+1+1 = 4\ (j = l), \qquad 1-1-1+1 = 0\ (j \neq l)$$

$$X_0'CX_0 = X_0'X_0 - \tfrac{1}{n}(X_0'\mathbf{1})(\mathbf{1}'X_0) = X_0'X_0 - \tfrac{1}{n}(n\bar{x})(n\bar{x}') = X_0'X_0 - n\bar{x}\bar{x}' \qquad \text{(Theorem 3)}$$

$$= \begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 8 & 16 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = 4I_2$$

$X_0'CX_0 = (CX_0)'(CX_0)$ is symmetric (Theorem 8).

Practice. The Exercises and Problems of the textbook, Chapter 1: products by entries, columns, rows and outer products; transposes and traces; diagonal and triangular matrices; the centering matrix and data matrices.